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A. Student (*remember to remove ID for the first draft and peer review!*)

October 17, 2026

Abstract

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1 Introduction

The Domain Name System (DNS) is a fundamental component of the Internet, responsible for translating human-readable domain names into IP addresses (Mockapetris, 1987). Most interactions with the internet require DNS resolution, which makes DNS queries a source of information for users' online activity. Originally, DNS queries have been transmitted without encryption, allowing all intermediaries with access to the datagram to observe which domains users access (Wicinski, 2021). Protocols such as DNS over HTTPS (DoH), DNS over TLS (DoT) and DNS over QUIC (DoQ) have been invented to provide a solution for this issue by encrypting the communication between clients and DNS servers (Hoffman and McManus, 2018; Hu et al., 2016; Huitema et al., 2022). However, while these protocols protect DNS queries from intermediaries in the network, the DNS server can still associate a query to the IP Address of the user that issued them.

Oblivious DNS over HTTPS (ODoH) addresses this issue by introducing an intermediary proxy that separates the client's identity from the query contents, forwarding encrypted requests to the target DNS server, which ensures that neither the proxy or the DNS server possess both the user's identity and the query contents (Singanamalla et al., 2020; Kinnear et al., 2022). Although this approach guarantees the privacy of the client's request end to end, it also prevents intermediary proxies from caching requests, since the query contents only get decrypted at the target resolver.

To overcome this issue, Niroula et al. (2026) developed Cacheable Oblivious DNS over HTTPS (CODoH) protocol. Rather than simply decrypting and caching DNS queries, CODoH combines trusted execution environments with techniques to prevent cache access patterns from revealing information. Its objective is to preserve the privacy properties of ODoH while partly recovering the performance benefits of DNS caching. However, these mechanisms also introduce overhead, raising questions about the performance of CODoH.

This report examines the architecture and privacy-enhancing mechanisms of CODoH.

2 Background

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Figure 1: An example privacy mechanism

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Theorem 1 (Incompleteness theorem, Gödel, 1931)**.**
If c be a given recursive, consistent class of formulae, then the propositional formula which states that c is consistent is not c-provable; in particular, the consistency of P is unprovable in P, it being assumed that P is consistent.

2.1 First background topic

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2.2 Second background topic

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3 Main content

Note that you *must* rename this section to indicate *what the content is*.

Header line	Column 2	Column 3
Content line	2	3
Another content line to showcase wrapping of table cells	2	3

Table 1: An example table

4 Another topic in the main content

Your "main content" can be split into as many sections as needed.

5 Related work

Here, you include short descriptions of similar work and explain *how* they are different from the main content you have presented.

Level of detail similar to an abstract.

Note that you may include the related work as a subsection in the Background Section 2.

6 Conclusion

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7 Use of generative digital tools

The use of AI tools must always be declared completely and transparently. For examples of how to do this, please see the DMI guidance: `https://dmi.unibas.ch/fileadmin/user_upload/dmi/Studium/Computer_Science/Diverses/Wegleitungen/Dokumente_2023/Guidance_on_AI_in_teaching_in_Computer_Science.pdf`.

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September 2023